STATISTICAL ARBITRAGE · บทที่ 6

Hurst Exponent & Stationarity Tests

$H < 0.5$ = Mean-Reverting · $H = 0.5$ = Random Walk · $H > 0.5$ = Trending

CORE IDEA — READ THIS FIRST

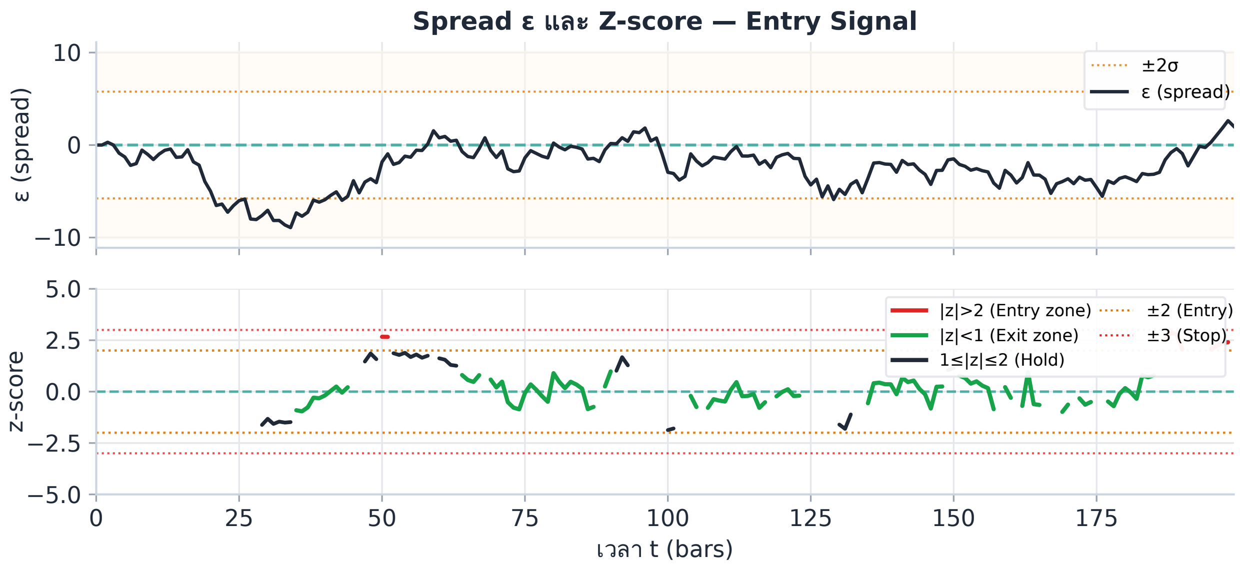
The **z-score** measures how far the current spread ε is from its mean, expressed in units of standard deviation. For statistical arbitrage, we **enter when |z| > 2** — the spread is unusually far from equilibrium — and **exit when |z| < 0.5** — the spread has reverted to normal. The z-score converts any spread into a dimensionless signal comparable across pairs and time.

$$z_t = \frac{\varepsilon_t - \mu_\varepsilon}{\sigma_\varepsilon}$$

`Entry: |z_t| > 2 · Exit: |z_t| < 0.5 · Stop: |z_t| > 3`

6.1 What H Measures — Three Regimes

The Hurst exponent was originally developed by Harold Hurst studying Nile River flood cycles. In finance, it quantifies whether a series has memory. Below, three synthetic time series illustrate the three regimes clearly.



Spread ε (top) with ±2σ bands and rolling z-score (bottom) colour-coded: RED = entry zone |z|>2, GREEN = exit zone |z|<1, NAVY = hold. Z-score computed on a 30-bar rolling window.

🏛️ **Practitioner Note — Rolling Window for μ and σ**

In live trading, **never use the full history** to compute μ_ε and σ_ε. Use a **rolling 30-day window** (≈ 720 hourly bars for crypto perps). This ensures the z-score adapts to recent regime shifts — e.g. if funding rates structurally change, the spread mean drifts and a full-history mean becomes stale within days. A rolling window keeps the denominator relevant and the z-score centred around zero. For very high-frequency pairs (5-minute bars), a 7-day window (2016 bars) is typical. Validate by checking that |mean(z)| < 0.1 on out-of-sample data.

⚠️ **Model Risk — Z-score Assumes Normal Distribution**

The z-score threshold of ±2 is calibrated to the Normal distribution, where P(|z|>2) ≈ 4.6%. In practice, spread residuals have **fat tails** (leptokurtosis) — the true probability of a 3σ event is often 2–5× higher than the Normal predicts. This means: (1) the entry threshold ±2 is triggered more frequently than expected — diluting the edge, and (2) stop-loss events at ±3 are more common — increasing tail risk. Mitigation: use a robust z-score with Median Absolute

Deviation (MAD) denominator, or widen entry to ±2.5 after observing the empirical exceedance rate on the pair's history.

ทำไม H < 0.5 ถึงสร้าง Edge

เมื่อ H < 0.5 การเคลื่อนไหวในอดีตทำนาย *ทิศตรงข้าม* ในอนาคต — ถ้า spread ขึ้น 1 standard deviation ใน bar นี้ bar ถัดไปน่าจะลง นี่คือแรงที่เราใช้ประโยชน์เมื่อ enter ที่ ±2σ และรอ reversion ยิ่ง H ต่ำ แรง mean-reversion ยิ่งแข็ง edge ต่อ trade ยิ่งสูง

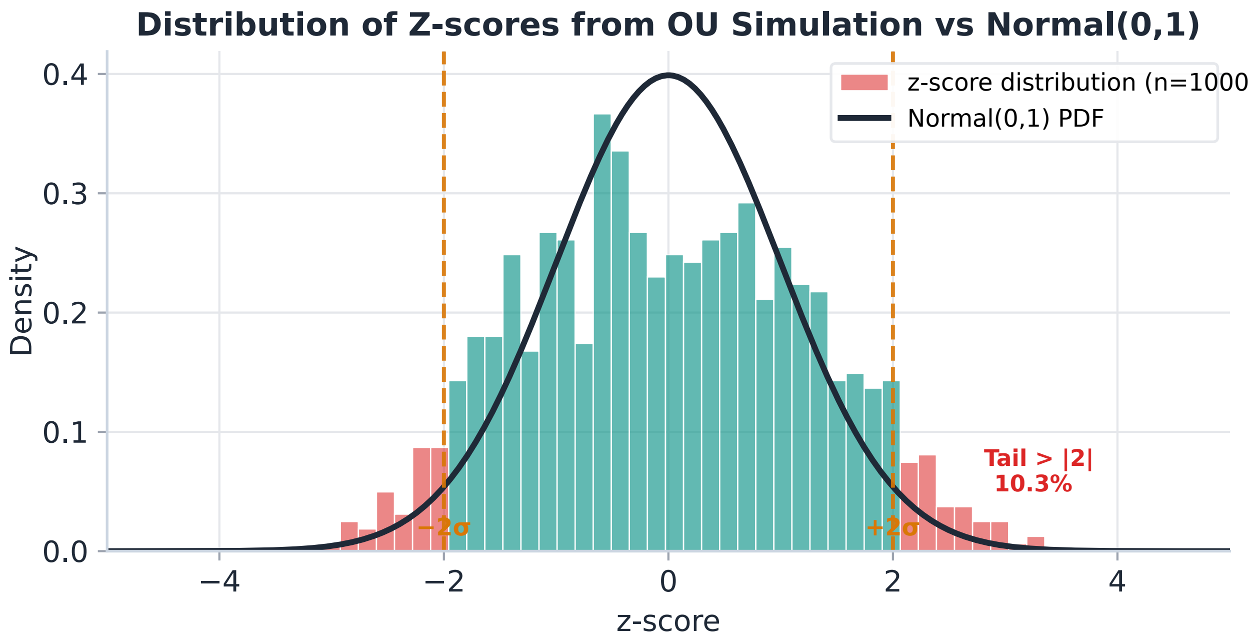
6.2 R/S Analysis — วิธีคำนวณ H

สัญชาตญาณ: R = range (สูงสุด – ต่ำสุดของ cumulative sum) วัดว่า series "แพร่กระจาย" ไปไกลแค่ไหน; S = std วัด volatility ตามปกติ อัตราส่วน R/S จึงบอกว่า series แพร่กระจายผิดปกติหรือเปล่า — random walk มี R/S ∝ √n แต่ mean-reverting series มี R/S เติบโตช้ากว่า slope H < 0.5 ในกราฟ log-log

วิธี **Rescaled Range (R/S)** คือ algorithm มาตรฐานในการ estimate H โดยวัดจากหลาย window size n:

1. Demean the series: y_i = x_i – mean(x_{1..n})
2. Compute cumulative sum: Y_k = Σ y_i for i=1..k
3. R(n) = max(Y) – min(Y) (range of cumulative sum)
4. S(n) = std(x_{1..n}) (standard deviation)
5. Rescaled range = R(n) / S(n)

Repeat for many window sizes n, then fit: **log(R/S) = H · log(n) + C**. The slope H is the Hurst exponent.



Distribution of z-scores from 1000 OU simulated observations vs Normal(0,1) PDF (navy). TEAL bars = simulated z-scores; RED bars = tails |z|>2. ±2σ shown as dashed AMBER lines.

6.3 ADF Test + KPSS Test — Double Confirmation

The Augmented Dickey-Fuller (ADF) and KPSS tests have **opposite null hypotheses**. Using both gives a robust verdict on stationarity.

Test	H_0 (Null)	Reject H_0 means	We want
ADF	Series is non-stationary (has unit root)	Series IS stationary	p-value < 0.05 (reject)
KPSS	Series IS stationary	Series is non-stationary	p-value > 0.05 (fail to reject)

Combined Interpretation — Four Cases

ADF result	KPSS result	Verdict	Trade?
Rejects (p < 0.05)	Fails to reject (p > 0.05)	**Strongly stationary**	Yes
Rejects (p < 0.05)	Rejects (p < 0.05)	Borderline / short memory	Caution
Fails to reject	Fails to reject (p > 0.05)	Inconclusive — collect more data / extend the sample window	No / wait
Fails to reject	Rejects (p < 0.05)	**Non-stationary**	Never

6.4 Interpretation Table — H Range to Trade Decision

H Range	Behavior	Trade Decision	Approx Half-Life
$H < 0.35$	Strongly mean-reverting	Trade aggressively, tighter bands	Very short (hours)
$0.35 \le H < 0.45$	Moderately mean-reverting	Trade normally, standard bands	Short to medium
$0.45 \le H < 0.55$	Near random walk — uncertain	Reduce size, require ADF confirm	Long or unreliable
$0.55 \le H < 0.65$	Weakly trending	Do not trade mean-reversion	N/A (trend instead)
$H \ge 0.65$	Strongly trending	Abort pair, find new one	N/A

Practical Note: Recalculate H Monthly

H is not stable over time. A pair that had $H = 0.33$ last month may shift to $H = 0.51$ after a structural break (exchange policy change, fee restructuring, or major liquidity event). Always recalculate H on a rolling 90-day window and alert if H crosses 0.45.

6.5 Pseudo-Code: hurst_rs()

```
function hurst_rs(series, min_window=10): # series: array of log-prices or spread values n_total = len(series) window_sizes = [10, 20, 40, 80, 160, 320] # geometric spacing log_n_list = [] log_rs_list = [] for n in window_sizes: if n >= n_total: continue rs_values = [] # Slide window (non-overlapping). ถ้าต้องการ H แม่นยำขึ้น # ใช้ overlapping windows (step=1) แต่ช้ากว่า O(n²) for start in range(0, n_total - n, n): chunk = series[start : start + n] demeaned = chunk - mean(chunk) cumsum = cumulative_sum(demeaned) R = max(cumsum) - min(cumsum) S = std(chunk) if S > 0: rs_values.append(R / S) if len(rs_values) > 0: log_n_list.append(log(n)) # ✅ FIX: mean(log(RS)) ไม่ใช่ log(mean(RS)) # log(mean(x)) ≠ mean(log(x)) โดย Jensen's inequality # การใช้ log(mean()) จะ bias H สูงขึ้น (overestimate mean-reversion) log_rs_list.append(mean([log(x) for x in rs_values])) # OLS regression: log(R/S) = H * log(n) + const H, const = ols_slope_intercept(log_n_list, log_rs_list) return H function check_stationarity(series): H = hurst_rs(series) adf_p = adf_test(series).p_value kpss_p = kpss_test(series).p_value tradeable = (H < 0.45) and (adf_p < 0.05) and (kpss_p > 0.05) return {"H": H, "adf_p": adf_p, "kpss_p": kpss_p, "tradeable": tradeable}
```

6.6 Confidence Interval สำหรับ H — Bootstrap Test

H = 0.31 กับ H = 0.35 **อาจแยกไม่ออกทางสถิติ** เพราะ R/S estimator มี noise โดยเฉพาะเมื่อ sample size น้อย การรายงาน H โดยไม่มี SE หรือ CI อาจทำให้ตัดสินใจ trade ผิด

Bootstrap CI สำหรับ H — แนวคิด

1. **Resample:** สุ่มดึง bootstrap sample ขนาดเท่า original (with replacement) ทำ B = 500 รอบ
2. **Compute H:** คำนวณ H จาก R/S analysis ทุก bootstrap sample → ได้ distribution $\{H_1, H_2, ..., H\_B\}$
3. **SE(H):** SE = std($\{H_1, ..., H\_B\}$)
4. **95% CI:** [H − 1.96·SE, H + 1.96·SE]

Rule: CI ต้อง **ทั้งหมดอยู่ต่ำกว่า 0.5** ถึงจะ trade ได้อย่างมั่นใจ ถ้า CI ทะลุ 0.5 ให้ลดขนาด position หรือรอ sample เพิ่ม

```
function hurst_bootstrap_ci(series, B=500, confidence=0.95): n = len(series)
h_samples = [] for _ in range(B): # Block bootstrap: สุ่ม block ขนาด sqrt(n) เพื่อ
รักษา autocorrelation block_size = int(sqrt(n)) resampled = [] while
len(resampled) < n: start = randint(0, n - block_size)
resampled.extend(series[start : start + block_size]) resampled = resampled[:n]
h_b = hurst_rs(resampled) h_samples.append(h_b) H_mean = mean(h_samples) H_se =
std(h_samples) alpha = 1 - confidence z_crit = 1.96 # 95% normal approximation
ci_low = H_mean - z_crit * H_se ci_high = H_mean + z_crit * H_se tradeable_ci =
ci_high < 0.50 # entire CI must be below 0.5 return {"H": H_mean, "SE": H_se,
"CI_95": (ci_low, ci_high), "tradeable_CI": tradeable_ci}
```

เหตุผลที่ใช้ Block Bootstrap แทน i.i.d. Bootstrap

Spread residuals มี autocorrelation (นั่นคือ mean-reverting behavior ที่เราวัด H) — ถ้า resample แบบ i.i.d. (สุ่มแต่ละจุดแยกกัน) จะทำลาย autocorrelation structure ทำให้ SE ของ H ต่ำเกินจริง (overconfident CI)

Block bootstrap ขนาด √n รักษา local autocorrelation ไว้ → SE ที่ได้จริงกว่า

H_point	SE (bootstrap)	95% CI	CI ต่ำกว่า 0.5?	ตัดสินใจ

0.31	0.03	[0.25, 0.37]	ใช่ ✓	Trade ได้ — confident
0.43	0.04	[0.35, 0.51]	ไม่ — CI ทะลุ 0.5	ลด size 50%, re-test next month
0.46	0.05	[0.36, 0.56]	ไม่ — CI ทะลุ 0.5	ไม่ trade จนกว่า H จะลง
0.38	0.02	[0.34, 0.42]	ใช่ ✓	Trade ได้ — บาง SE → น่าเชื่อถือ

ตัวอย่างจริง: Bybit BTC Perp ↔ Lighter BTC Perp

We compute H on two series — the spread ε and raw BTC price — over the last 90 days of hourly data:

Series	H (R/S)	Interpretation	ADF p-value	KPSS p-value	Verdict
Spread ε (Bybit – 1.003·Lighter)	**0.31**	Strongly mean-reverting	0.0003	0.18	**Trade ✓**
Raw BTC log-price (Bybit)	**0.52**	Near random walk	0.41	0.02	**Do not trade**

The spread has H = 0.31 — deeply in mean-reverting territory. ADF rejects non-stationarity (p = 0.0003) AND KPSS fails to reject stationarity (p = 0.18). This is the strongest possible confirmation. The raw BTC price, by contrast, shows H = 0.52 — a near-perfect random walk, exactly as theory predicts.

```
Estimated half-life from H = 0.31: → Hurst-implied: ~4–8 hours (tight, fast reversion) → Confirmed by OU fit: κ = 2.8/day → half-life ≈ 5.9 hours
```

โจทย์ 6.1 — แปลความหมาย H = 0.43

A new pair (Binance BTC Perp ↔ OKX BTC Perp) produces H = 0.43 from the R/S analysis. ADF p = 0.032, KPSS p = 0.08.

Q: Is this spread mean-reverting? What trading approach do you take?

▶ ดูคำตอบ

โจทย์ 6.2 — ผลทดสอบขัดแย้งกัน

For a candidate spread: ADF p = 0.08 (fails to reject), KPSS p = 0.03 (rejects stationarity).

Q: Interpret this combined result. Should you trade this spread?

▶ ดูคำตอบ

Ch.6 Hurst Exponent & Stationarity Tests | ต่อไป → **Ch.7: GARCH on Residuals**